Simply Supported Steel Beam — quick check

1. Input Parameters

Symbol	Value	Unit	Description
$beam_{h1}$	1		beam_h1
$Span_{L}$	6	m	Span L (m)
$Line_{load,\,w}$	12	kN/m	Line load w (kN/m)
$Section_{modulus,\,Wel}$	324	cm3	Section modulus Wel (cm3)
$Yield_{fy}$	355	MPa	Yield fy (MPa)
$factors_{b2}$	1		factors_b2

2. Calculation

Moment M (kNm)

$$Moment_{M} = \frac{Line_{load,\,w} \cdot Span_{L}^{2}}{8}$$
$$= \frac{12 \cdot 6^{2}}{8}$$
$$= \mathbf{54}\ \mathrm{kNm}$$

Bending stress sigma (MPa)

$$Bending_{stress,\,\sigma} = Moment_{M} \cdot 1000 \cdot \frac{1}{Section_{modulus,\,Wel}} \cdot factors_{b2}$$
$$= 54 \cdot 1000 \cdot \frac{1}{324} \cdot 1$$
$$= \mathbf{166.7}\ \mathrm{MPa}$$

3. Verification

OK $Bending_{stress,\,\sigma} = 166.7\ \mathrm{MPa} \ \leq \ Yield_{fy} = 355\ \mathrm{MPa}$